

DS-GA 1011: Natural Language Processing

Intermediate Check-in

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1 Project Description

The main goal of this project is to evaluate how different forms of reasoning-based supervision affect the performance of retrieval-augmented generation (RAG) systems on multi-hop question answering. Specifically, we compare three RAG variants—**Vanilla RAG**, **Self-RAG**, and **InstructRAG**—under a unified fine-tuning and evaluation framework. Building on our initial proposal, which originally included RAFT as a third method, we transitioned to InstructRAG to maintain methodological comparability while avoiding the heavy retriever-generator co-training that RAFT requires. All models use the same backbone language model and retriever, and are trained via **supervised fine-tuning (SFT)** on the same dataset split to ensure consistent optimization. For our backbone model, we have shifted from Llama 3.1 8B to **Llama 2 7B** to align with the original Self-RAG implementation and ensure direct reproducibility of baseline results. The key difference across variants lies in the **type of reasoning supervision** applied to the generator: Vanilla RAG uses standard question-answer supervision; Self-RAG augments this with reflection tokens that enable adaptive retrieval and iterative reasoning; and InstructRAG employs rationale-based instruction tuning that improves evidence utilization and robustness to retrieval noise. All models are evaluated on the **HotpotQA** distractor benchmark using **Exact Match (EM)** and **F1** scores to measure factual correctness and token-level overlap. This design isolates the contribution of different reasoning signals—reflection-driven versus instruction-driven—while controlling for data, retriever, and training conditions.

2 Achieved Milestones

Table 1: RAG Performance on HotpotQA Dev Set (500 Examples)

Model	Exact Match (EM)	F1 Score
Vanilla RAG	6.40%	14.03%
Self-RAG	43.80%	28.78%
InstructRAG	7.80%	15.20%

For this midterm evaluation, we report results on the HotpotQA development set (500 examples) as we have not yet completed runs on the full 90,447-example training dataset. Where applicable, all three models share common baseline parameters (`temperature` = 0, `top_k` = 5, `top_p` = 1), though Self-RAG and InstructRAG include additional model-specific hyperparameters inherent to their architectures. Additionally, we use LLaMA-2-7B Base for the vanilla RAG setup and the instruction-tuned LLaMA-2-7B-Chat variant for InstructRAG.

2.1 Vanilla RAG Implementation

Our vanilla RAG implementation uses meta-llama/Llama-2-7b-hf as the backbone language model. The system retrieves passages from pre-computed document contexts and concatenates them with the input question, then employs greedy decoding to generate answers with a maximum length of 15 tokens to match HotpotQA’s short-answer format.

We evaluate on the HotpotQA development set (500 examples) using vLLM for efficient batch inference, measuring Exact Match (EM) and token-level F1 scores. Vanilla RAG achieves an EM of 6.40% and F1 of 14.03%. This relatively low performance reflects the inherent limitations of vanilla RAG on multi-hop reasoning tasks: simple concatenation of retrieved passages lacks sophisticated reasoning mechanisms to integrate information across multiple hops, and the passive retrieval process can introduce noise from irrelevant documents.

The F1 score roughly doubling the EM indicates that the model captures partial information but rarely produces exact matches, suggesting it understands relevant context but struggles with precise answer extraction. This baseline establishes the performance floor against which we will compare Self-RAG and InstructRAG.

2.2 Self-RAG Implementation

We evaluate the pre-trained `selfrag/selfrag_llama2.7b` model, an extension of Llama 2 7B that introduces reflection tokens enabling adaptive retrieval and self-assessment during generation (Asai et al., 2023). In contrast to vanilla RAG’s passive retrieval pipeline, Self-RAG actively decides when to retrieve and how to assess the relevance, groundedness, and utility of the retrieved evidence.

Using vLLM (v0.11.0) with greedy decoding, we retrieve the top 10 passages per question and rank them using reflection-token scores with weights $w_{\text{rel}} = 1.0$, $w_{\text{sup}} = 1.0$, and $w_{\text{use}} = 0.5$. Following the baseline configuration in Asai et al. (2023), we set the retrieval threshold to 0.2. Due to T4 GPU memory limitations (15 GB), we also reduce the maximum context length from 2048 to 1472 tokens.

As shown in our results, Self-RAG achieves 43.80% EM and 28.78% F1—representing a $6.8\times$ improvement over vanilla RAG. Its high retrieval rate (99.2%) indicates that Self-RAG consistently recognizes HotpotQA queries as requiring external knowledge, with 81.0% of answers fully supported by retrieved documents. To further explore Self-RAG’s behavior, we conduct threshold and weight ablation studies. Threshold ablations (Table 2) show that thresholds in the 0.1–0.3 range yield similar performance (44.0–44.2% EM and 99.2% retrieval), whereas thresholds ≥ 0.5 suppress retrieval entirely, reducing performance to 19.8% EM—demonstrating a +24.4-point boost attributable to retrieval. This binary behavior reveals that Self-RAG’s retrieval probabilities cluster tightly around 0.4–0.6 for HotpotQA, indicating that the model has learned to recognize multi-hop reasoning questions as inherently requiring external knowledge. Weight ablations (Table 3) show minimal variation (43.60–43.80% EM), suggesting that Self-RAG’s weighting scheme is already well calibrated. The insensitivity to weight changes indicates that passage ranking is robust, with relevance and groundedness serving as the primary drivers of performance.

Table 2: Self-RAG Threshold Ablation Study (500 Examples)

Threshold	EM (%)	F1 (%)	Retrieval Rate (%)
0.1	44.20	28.82	99.2
0.2	44.00	28.85	99.2
0.3	44.00	28.65	99.2
0.5	19.80	13.79	0.0
0.7	19.80	13.79	0.0
0.9	19.80	13.79	0.0

Table 3: Self-RAG Weight Ablation Study (Threshold=0.2, 500 Examples)

Configuration F1 (%)	w_{rel}	w_{sup}	w_{use}	EM (%)
Baseline 28.78	1.0	1.0	0.5	43.80
No Utility 29.00	1.0	1.0	0.0	43.60
Emphasize Groundedness 29.04	1.0	2.0	0.0	43.80

2.3 Transition from RAFT to InstructRAG

Our initial proposal selected **RAFT** as the second RAG variant for comparison, as it also fine-tunes the generator to reason more effectively over retrieved evidence. However, a closer examination of RAFT’s training paradigm revealed that it is primarily a *training methodology* rather than a fixed model configuration. Implementing RAFT would require constructing specialized training data (question, retrieved passages, oracle snippets) and fully fine-tuning the backbone language model under this regime (Zhang et al., 2024). This raised concerns about comparability since training a model from scratch would introduce confounding factors and make it difficult to isolate differences attributable solely to retrieval-generation design.

To preserve methodological alignment with **vanilla RAG** and **Self-RAG**, we adopted **InstructRAG** instead. Like Self-RAG, InstructRAG maintains the retrieval-generation structure and adapts the generator through

instruction-based fine-tuning on retrieval-augmented data, allowing for a controlled, directly comparable experimental setup (Wei et al., 2024). This substitution retains the spirit of evaluating fine-tuned RAG variants while remaining feasible within our available compute budget. We revise the prompt and have successfully run the training-free version of InstructRAG pipeline using the LLaMA-2-7B-chat backbone. So far, we stay on chat version because it follows the original prompt format and LLM version of InstructRAG. After the midterm, we will try to switch to pure LLaMA-2-7B to align the configurations of the other RAGs exactly, and compare the prompts used in different RAGs to see if the performance rely on the prompt and LLM version heavily.

3 Literature Survey

Retrieval-Augmented Generation (RAG) addresses fundamental limitations of Large Language Models by integrating external knowledge retrieval with text generation (Lewis et al., 2020). While vanilla RAG achieved significant improvements on open-domain question answering by treating retrieved documents as latent variables that condition generation, it revealed critical limitations on complex tasks like multi-hop reasoning: indiscriminate retrieval regardless of necessity, inability to filter irrelevant passages, and lack of mechanisms to synthesize information across multiple documents (Asai et al., 2023; Zhang et al., 2024). These shortcomings are particularly evident on benchmarks like HotpotQA (Yang et al., 2018), where questions require reasoning chains across multiple evidence sources and models must distinguish relevant passages from distractors.

Recent work has explored different paradigms for improving how models leverage retrieved knowledge. Self-RAG introduces adaptive retrieval through reflection tokens that enable models to decide when to retrieve, assess passage relevance (IsREL), verify evidence support (IsSUP), and evaluate output usefulness (IsUSE) (Asai et al., 2023). This reflection-driven approach provides inference-time controllability and demonstrated significant gains in factuality and citation accuracy. In contrast, InstructRAG employs rationale-based instruction tuning, where models are fine-tuned with self-synthesized explanations that improve evidence utilization and robustness to retrieval noise (Wei et al., 2024). Unlike Self-RAG’s token-level reflection mechanism, InstructRAG’s approach teaches reasoning through explicit rationale generation during training, offering a complementary strategy for enhancing RAG performance.

Despite these advances, there remains limited empirical understanding of how reflection-driven versus instruction-driven reasoning supervision affects performance under controlled conditions. Our work provides the first direct comparison of vanilla RAG, Self-RAG, and InstructRAG using identical backbone models (Llama 2 7B), retrievers, training data, and evaluation protocols on the HotpotQA distractor benchmark. By isolating the contribution of different reasoning signals while controlling for confounding factors, we address key questions about which supervision strategy is most effective for multi-hop question answering and under what conditions each approach succeeds or fails.

4 Member Contribution

Ghina Al Shdaifat established the foundational infrastructure for the project, implementing the core vanilla RAG pipeline including the retrieval mechanism, prompt templates, and evaluation framework. She is currently leading the Self-RAG implementation, working on integrating reflection token generation and adaptive retrieval mechanisms.

Judy Yang conducted the vanilla RAG experiments, performed the evaluation analysis, and produced the baseline results. She also contributed to the experimental design and evaluation metrics setup.

Shengduo Li, Huizhen Jin, and Yixuan Wang are jointly working on the InstructRAG implementation. They are responsible for adapting the InstructRAG evaluation pipeline, refining the prompt templates, aligning the evaluation protocol with the Self-RAG setup, and conducting the evaluation and analysis of results.

All team members contributed equally to the literature review, project proposal, and this intermediate check-in report. Writing responsibilities were distributed collaboratively, with iterative review and editing from all members.

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